Bhadrinathan A

Lead Engineer | Full-Stack Developer

LinkedIn | bhad3.in

Phone: **+91 7010198469**
Email: **bhadrinathan28@gmail.com**
GitHub: Bhadri01

Profile Summary

- **AI & Full-Stack Lead Engineer** with 4+ years of experience engineering real-time architectures, distributed systems, and AI applications.
- Specialized in Python (FastAPI), Rust (Axum), MERN stack, and enterprise generative AI integration.
- Architected core EdTech platforms at BloomSkillTech; mentored 1,000+ engineers across institutions.
- Proficient in RAG pipelines, LangChain, MCP, LLM APIs, and high-performance Rust tooling.

Technical Skills

Programming Languages	**Python**, **JavaScript**, **TypeScript**, **Rust**.
Backend & APIs	**FastAPI**, **SQLAlchemy**, **Axum**, REST APIs, SSE, WebSockets, **Celery**
Web & Mobile Frameworks	**React**, **Next.js**, Express, **Node.js** (MERN), **React Native**.
Databases & Messaging	**PostgreSQL**, **MongoDB**, **Redis/Valkey**, **TimescaleDB**, Alembic, MinIO/S3.
Infrastructure & Quality	**Linux**, WireGuard, **Traefik/Nginx**, **Pytest**.
DevOps & Tools	**Docker**, VCS (GitHub, GitLab), Jira, **GitHub Actions**, **CI/CD**.
AI Learning & Applied	**RAG**, **pgvector**, **Qdrant**, **LangChain**, **LangGraph**, **MCP**, LLM APIs.
AI-Assisted Dev	**Claude**, Codex, **GitHub Copilot**, Antigravity

Work Experience

Senior Software Engineer - EdTech Platform | BloomSkillTech, Salem | Jan 2025 – Present

- Architected end-to-end multi-tenant **EdTech platform** for high-concurrency workloads.
- Led **React** and **FastAPI/PostgreSQL** engineering integrated with **LLM APIs and RAG**.
- Established automated **CI/CD pipelines**, test suites, and deployment protocols.
- Engineered multi-agent pipelines using **LangChain, LangGraph, and MCP**.
- Optimized RAG pipeline indexing and vector queries, cutting response latency by 40% under high concurrency.
- Mentored 6+ engineers, driving code reviews, architectural standards, and team-wide AI tooling adoption.

Software Engineer | BloomSkillTech, Salem, | May 2023 – Dec 2024

- **Deployed cloud labs platform** with interactive coding environments over a **secure VPN architecture**.
- Created containerized environments using **Docker** to dynamically isolate and scale user workspaces.
- Built **FastAPI** microservices to manage container lifecycles, resources, and real-time sessions.
- Engineered Redis caching and asynchronous job queues, boosting API throughput by 50% and reducing server overhead.

Associate Software Engineer | BloomSkillTech, Salem | Dec 2022 – May 2023

- **Engineered a law enforcement CRM** with case timelines, full-text record search, and tracking.
- **Developed React UI, Go REST APIs, PostgreSQL schema**, and containerized microservices with Docker.

Software Engineer Intern | Salem City Cyber Crime Police | Aug 2021 - Dec 2022

- Developed a **website incorporating the station's database** and continued **software maintenance and support** until joining BloomSkillTech.
- Received a **Certificate of Appreciation from Salem City Police in December 2021** for website development contributions.

Open Source & Projects

- **fastapi-querybuilder** | Python, FastAPI | PyPI: Optimized query compilation performance; surpassed 18,000+ PyPI downloads.
- **ZeroCode** | Rust: Boosted sandboxed execution engine performance by 70% over traditional tools with an integrated caching system and hardened security against 130+ adversarial test cases.

Achievements

- **Rated Chess Player** holding a 1612 official FIDE Rating.
- **3rd Place Winner** in Yukthi CTF competition hosted by Selfmade Ninja Academy.

Education

- **B.E. Electronics and Communication Engineering** | 2019 – 2023, Muthayammal Engineering College, Namakkal.